

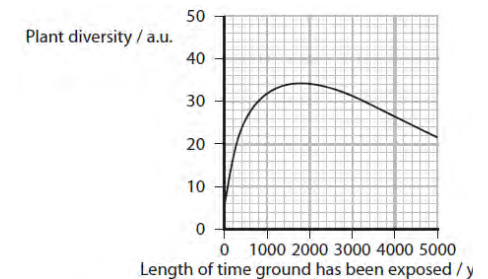
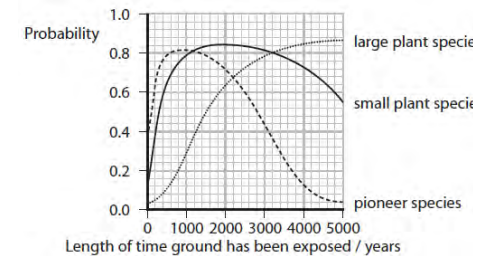
Question number	Answer	Additional guidance	Mark
4(a)(i)	{0.29 / 0.3} metres per year metres yr⁻¹ m yr⁻¹ m per y metres / year (1)	ACCEPT with or without - sign DO NOT ACCEPT m per yr ⁻¹ m / yr ⁻¹ m ⁻¹	(1)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	An explanation that includes the following points: - because of global warming (being greater) (1) - so {temperatures were higher / increase in temperature was greater} (1) - therefore melting (more) {ice / glacier} (1)	NB converse if talking about 1980 to 1985 ACCEPT example of something that causes increase in global warming e.g. more fossil fuels burnt, more greenhouse gases IGNORE climate change	(3)

Question number	Answer	Mark
4(b)	The only correct answer is B <i>A is incorrect because pioneer species will not grow on the ice</i> <i>C is incorrect because pioneer species grow on newly exposed ground</i> <i>D is incorrect because pioneer species grow on newly exposed ground</i>	(1)

Question number	Answer	Mark
4(c)(i)	The only correct answer is D <i>A is incorrect because anthropogenic is the effect of humans</i> <i>C is incorrect because evolution is the change in inheritable characteristics over time</i> <i>D is incorrect because speciation is the formation of a new species</i>	(1)

Question number	Answer
c)(ii)	Indicative content: Graph 1 possible explanations : • conditions and organisms vary with distance from glacier due to succession • pioneer species arrive first because {conditions hostile / no soil / dry conditions} • pioneer species die and improve the mineral / humus content in the soil • these new conditions support small plants • as small plants arrive, they compete with pioneer species for resources • as small plants die they improve the soil mineral content / depth • which will support larger plants • which then compete with small plants for resources • until a climax community is reached • but this is not shown on the graph as the numbers are still changing Graph 2 possible explanations : • biodiversity is low when only pioneer species present • increases as small plants arrive • increases further as larger plants begin to arrive • decreases as more larger plants grow • as fewer plant species can be supported • may decrease due to habitat loss Carbon:nitrogen table possible explanations : • ratio increases as more carbon added to soil / nitrogen is removed from the soil • early increase corresponds with increase in pioneer species • as dead plants decompose adding carbon / humus to the soil • further increase as small plants arrive and die • animals will be attracted to area with increase in plants species • as plants provide food and shelter • animal excrement and decomposing dead animals will add nitrogen to the soil • decrease in ratio corresponds with increase in number of large plant species • as more nitrogen because of leaf litter / nitrogen-fixing bacteria / nitrifying bacteria • as less carbon due to more decomposition / respiration by decomposers



Length of time that the ground has been exposed / years	Carbon : nitrogen ratio in the soil
< 200	9.8
2 000	11.6
3 500	10.7
7 200	10.3

			Additional guidance
evel 0	0	No awardable content	
evel 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	Simple description /explanation of data with no real attempt of an explanation 1 mark = some description of the data 2 marks = simple explanation of one of: succession or biodiversity or table
evel 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	Some explanation given using the information given / own knowledge 3 marks = simple explanation of two of: succession or biodiversity or table 4 marks = more detailed explanation one of: succession or biodiversity or table
evel 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	Detailed explanation given using the information given / own knowledge 5 marks = detailed explanation of two of: succession or biodiversity or table wit some ref to data 6 marks = detailed explanation of succession and biodiversity that also demonstrates an understanding of C : N ratio

